5

Curved Exponential Families

Curved exponential families may arise when the parameters of an exponential family satisfy constraints. For these families the minimal sufficient statistic may not be complete, and UMVU estimation may not be possible. Curved exponential families arise naturally with data from sequential experiments, considered in Section 5.2, and Section 5.3 considers applications to contingency table analysis.

5.1 Constrained Families

Let $\mathcal{P} = \{P_\eta : \eta \in \Xi\}$ be a full rank s-parameter canonical exponential family with complete sufficient statistic T. Consider a submodel $\mathcal{P}_0$ parameterized by $\theta \in \Omega$ with $\tilde{\eta}(\theta)$ the value for the canonical parameter associated with θ. So

$$\mathcal{P}_0 = \{P_{\tilde{\eta}(\theta)} : \theta \in \Omega\}.$$

Often $\tilde{\eta} : \Omega \to \Xi$ is one-to-one and onto. In this case $\mathcal{P}_0 = \mathcal{P}$ and the choice of parameter, θ or η, is dictated primarily by convenience. Curved exponential families may arise when $\mathcal{P}_0$ is a strict subset of $\mathcal{P}$, generally with $\Omega \subset \mathbb{R}^r$ and $r < s$. Here are two possibilities.

1. Points η in the range of $\tilde{\eta}$, $\tilde{\eta}(\Omega) = \{\tilde{\eta}(\theta) : \theta \in \Omega\}$, satisfy a nontrivial linear constraint. In this case, $\mathcal{P}_0$ will be a q-parameter exponential family for some $q < s$. The statistic T will still be sufficient, but will not be minimal sufficient.
2. The points η in $\tilde{\eta}(\Omega)$ do not satisfy a linear constraint. In this case, $\mathcal{P}_0$ is called a *curved exponential family*. Here T will be minimal sufficient (see Example 3.12), but may not be complete.

Example 5.1. Joint distributions for a sample from $N(\mu, \sigma^2)$ form a two-parameter exponential family with canonical parameter

R.W. Keener, *Theoretical Statistics: Topics for a Core Course*, Springer Texts in Statistics, DOI 10.1007/978-0-387-93839-4_5, © Springer Science+Business Media, LLC 2010

$$\eta = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right)$$

and complete sufficient statistic

$$T = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right).$$

(See Example 2.3.) If μ and σ^2 are equal and we let θ denote the common value, then our subfamily will consist of joint distributions for a sample from $N(\theta, \theta)$, $\theta > 0$. Then

$$\tilde{\eta}(\theta) = \left(1, -\frac{1}{2\theta} \right),$$

and the range of $\tilde{\eta}$ is the half-line indicated in Figure 5.1. Because points η in $\tilde{\eta}(\Omega)$ satisfy the linear constraint $\eta_1 = 1$, the subfamily should be exponential with less than two parameters. This is easy to check; the joint densities form a full rank one-parameter exponential family with $\sum_{i=1}^{n} X_i^2$ as the canonical complete sufficient statistic.

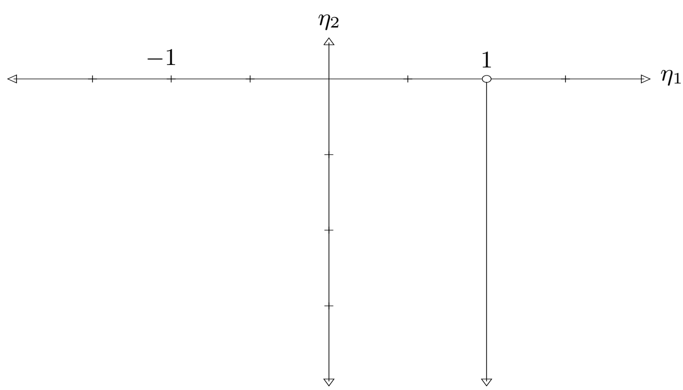

Fig. 5.1. Range of $\tilde{\eta}(\theta) = \big(1, -1/(2\theta)\big)$.

Suppose instead $\sigma = |\mu|$, so the subfamily will be joint distributions for a sample from $N(\theta, \theta^2)$, $\theta \in \mathbb{R}$. In this case

$$\tilde{\eta}(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2} \right).$$

Now the range space $\tilde{\eta}(\Omega)$ is the parabola in Figure 5.2. Points in this range space do not satisfy a linear constraint, so in this case we have a curved exponential family and T is minimal sufficient. Because

$$E_\theta T_1^2 = (E_\theta T_1)^2 + \mathrm{Var}_\theta(T_1) = n^2\theta^2 + n\theta^2,$$

and

$$E_\theta T_2 = nE_\theta X_i^2 = n\big((E_\theta X_i)^2 + \mathrm{Var}_\theta(X_i)\big) = 2n\theta^2,$$

we have

$$E_\theta\big(2T_1^2 - (n+1)T_2\big) = 0, \qquad \theta \in \mathbb{R}.$$

Thus $g(T) = 2T_1^2 - (n+1)T_2$ has zero mean regardless the value of θ. Inasmuch as $g(T)$ is not zero (unless $n = 1$), T is not complete.

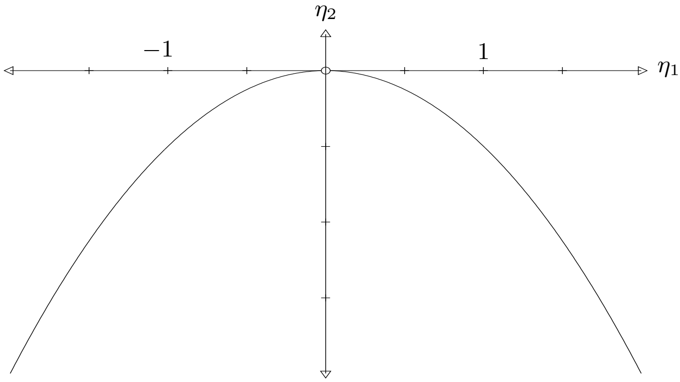

Fig. 5.2. Range of $\tilde{\eta}(\theta) = \big(1/\theta, -1/(2\theta^2)\big)$.

Example 5.2. If our data consist of two independent random samples, $X_1, \dots, X_m$ from $N(\mu_x, \sigma_x^2)$ and $Y_1, \dots, Y_n$ from $N(\mu_y, \sigma_y^2)$, then the joint distributions form a four-parameter exponential family indexed by $\theta = (\mu_x, \mu_y, \sigma_x^2, \sigma_y^2)$. A canonical sufficient statistic for the family is

$$T = \left(\sum_{i=1}^m X_i, \sum_{j=1}^n Y_j, \sum_{i=1}^m X_i^2, \sum_{j=1}^n Y_j^2\right),$$

and the canonical parameter is

$$\eta = \left(\frac{\mu_x}{\sigma_x^2}, \frac{\mu_y}{\sigma_y^2}, -\frac{1}{2\sigma_x^2}, -\frac{1}{2\sigma_y^2}\right).$$

By (4.3) and (4.4), an equivalent statistic would be $(\overline{X}, \overline{Y}, S_x^2, S_y^2)$, where $S_x^2 = \sum_{i=1}^m (X_i - \overline{X})^2/(m-1)$ and $S_y^2 = \sum_{i=1}^n (Y_i - \overline{Y})^2/(n-1)$. Results from Section 4.3 provide UMVU estimates for μ_x, μ_y, σ_x^r, σ_y^r, etc.

If the variances for the two samples agree, $\sigma_x^2 = \sigma_y^2 = \sigma^2$, then η satisfies the linear constraint $\eta_3 = \eta_4$. In this case the joint distributions form a three-parameter exponential family with complete sufficient statistic $(T_1, T_2, T_3 + T_4)$. An equivalent sufficient statistic here is $(\overline{X}, \overline{Y}, S_p^2)$, where

$$S_p^2 = \frac{\sum_{i=1}^m (X_i - \overline{X})^2 + \sum_{i=1}^n (Y_i - \overline{Y})^2}{n+m-2} = \frac{(m-1)S_x^2 + (n-1)S_y^2}{n+m-2},$$

called the *pooled sample variance*. Again the equivalence follows easily from (4.3) and (4.4). Also, because S_x^2 and S_y^2 are independent, from the definition of the chi-square distribution and (4.9),

$$\frac{(n+m-2)S_p^2}{\sigma^2} \sim \chi^2_{n+m-2}.$$

Again, results from Section 4.3 provide UMVU estimates for various parameters of interest.

Another subfamily arises if the means for the two samples are the same, $\mu_x = \mu_y$. In this case the joint distributions form a curved exponential family, and T or $(\overline{X}, \overline{Y}, S_x^2, S_y^2)$ are minimal sufficient. In this case these statistics are not complete because $E(\overline{X} - \overline{Y}) = 0$ for all distributions in the subfamily.

5.2 Sequential Experiments

The protocol for an experiment is *sequential* if the data are observed as they are collected, and the information from the observations influences how the experiment is performed. For instance, the decision whether to terminate a study at some stage or continue collecting more data might be based on prior observations. Or, in allocation problems sampling from two or more populations, the choice of population sampled at a given stage could depend on prior data.

There are two major reasons why a sequential experiment might be preferred over a classical experiment. A sequential experiment may be more efficient. Here efficiency gains might be quantified as a reduction in decision theoretic risk, with costs for running the experiment added to the usual loss function. There are also situations in which certain objectives can only be met with a sequential experiment. Here is one example.

Example 5.3. Estimating a Population Size. Consider a lake (or some other population) with M fish. Here M is considered an unknown parameter, and the goal of the experiment is to estimate M. Data to estimate M are obtained from a "capture–recapture" experiment. This experiment has two phases. First, k fish are sampled from the lake and tagged so they can be identified. These fish are then returned to the lake. At the second stage, fish are sampled at random from the lake. Note that during this phase a sampled fish is tagged with probability $\theta = k/M$. (Actually, there is an assumption here that at the second stage tagged and untagged fish are equally likely to be captured; this premise seems suspect for real fish.) In terms of θ,

$$M = k/\theta,$$

and so the inferential goal is basically to estimate $1/\theta$. Information from the second stage of this experiment can be coded using Bernoulli variables $X_1, \dots, X_N$, where N denotes the sample size, and X_i is one if the ith fish is tagged or zero if the ith fish is not tagged.

In mathematical terms we have a situation in which potential data $X_1, X_2, \dots$ are i.i.d. Bernoulli variables with success probability θ. If the sample size is fixed, $N = n$, then our data have joint density

$$\prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{T(x)}(1-\theta)^{n-T(x)},$$

where $T(x) = x_1 + \cdots + x_n$. These densities form an exponential family with T as a sufficient statistic. Because T has a binomial distribution with mean $n\theta$, T/n is unbiased for θ and hence UMVU. But there can be no unbiased estimate of $1/\theta$ because

$$E_\theta \delta(T) \le \max_{0 \le k \le n} \delta(k),$$

which is less than $1/\theta$ once θ is sufficiently small. Note that if θ is much smaller than $1/n$, then T will be zero with probability close to one. The real problem here is that when $T = 0$ we cannot infer much about the relative size of θ from our data.

Inverse binomial sampling avoids the problem just noted by continued sampling until m of the X_i equal one. The number of observations N is now a random variable. Also, this is a sequential experiment because the decision to stop sampling is based on observed data.

Intuitively, data from inverse binomial sampling would be the list

$$X = (X_1, \dots, X_N).$$

There is a bit of a technical problem here: this list is not a random vector because the number of entries N is random. The most natural way around this trouble involves a more advanced notion of "data" in which the information from an experiment is viewed as the σ-field of events that can be resolved from the experiment. Here this σ-field would include events such as $\{T = k\}$ or $\{N = 7\}$, but would preclude events such as $\{X_{N+2} = 0\}$. See Chapter 20 for a discussion of this approach. Fortunately, in this example we can avoid these technical issues in the following fashion. Let Y_1 be the number of zeros in the list X before the first one, and let Y_i be the number of zeros between the $(i-1)$st and ith one, $i = 2, \dots, m$. Note that the list X can be recovered from $Y = (Y_1, \dots, Y_m)$. If, for instance, $Y = (2, 0, 1)$, then X must be $(0, 0, 1, 1, 0, 1)$. The variables $Y_1, \dots, Y_m$ are i.i.d. with

$$\begin{aligned} P_\theta(Y_i = y) &= P(X_1 = 0, \dots, X_y = 0, X_{y+1} = 1) \\ &= (1-\theta)^y \theta \\ &= \exp\bigl(y \log(1-\theta) + \log \theta\bigr), \qquad y = 0, 1, \dots. \end{aligned}$$

This is the mass function for the geometric distribution. It is a one-parameter exponential family with canonical parameter $\eta = \log(1-\theta)$ and

$$A(\eta) = -\log\theta = -\log(1-e^\eta).$$

Thus

$$E_\theta Y_i = A'(\eta) = \frac{e^\eta}{1-e^\eta} = \frac{1-\theta}{\theta}.$$

The family of joint distributions of $Y_1, \dots, Y_m$ has $T = \sum_{i=1}^m Y_i = N - m$ as a complete sufficient statistic. The statistic T counts the number of failures before the mth success and has the negative binomial distribution with mass function

$$P(T=t) = \binom{m+t-1}{m-1}\theta^m(1-\theta)^t, \qquad t = 0, 1, \dots.$$

Inasmuch as

$$E_\theta T = mE_\theta Y_i = \frac{m}{\theta} - m,$$
$$\frac{T+m}{m} = \frac{N}{m}$$

is UMVU for $1/\theta$.

The following result gives densities for a sequential experiment in which data $X_1, X_2, \dots$ are observed until a stopping time N. This stopping time is allowed to depend on the data, but clairvoyance is prohibited. Formally, this is accomplished by insisting that

$$\{N=n\} = \bigl\{(X_1,\dots,X_n) \in A_n\bigr\}, \qquad n = 1, 2, \dots,$$

for some sequence of sets $A_1, A_2, \dots$.

Theorem 5.4. *Suppose $X_1, X_2, \dots$ are i.i.d. with common marginal density f_θ, $\theta \in \Omega$. If $P_\theta(N < \infty) = 1$ for all $\theta \in \Omega$, then the total data, viewed informally*[1] *as $(N, X_1, \dots, X_N)$, have joint density*

$$\prod_{i=1}^n f_\theta(x_i). \tag{5.1}$$

When f_θ comes from an exponential family, so that

$$f_\theta(x) = e^{\eta(\theta)\cdot T(x) - B(\theta)}h(x),$$

[1] One way to be more precise is to view the information from the observed data as a σ-field. This approach is developed in Section 20.2, and Theorem 20.6 (Wald's fundamental identity) from this section is the mathematical basis for the theorem here.

then the joint density is

$$\exp\left[\eta(\theta)\cdot\sum_{i=1}^{n}T(x_i)-nB(\theta)\right]\prod_{i=1}^{n}h(x_i). \tag{5.2}$$

These densities form an exponential family with canonical parameters $\eta_1(\theta)$, $\dots$, $\eta_s(\theta)$, *and* $-B(\theta)$, *and sufficient statistic* $\left(\sum_{i=1}^{N}T(X_i), N\right)$.

By (5.1), the likelihood for the sequential experiment is the same as the likelihood that would be used ignoring the optional stopping and treating N as a fixed constant. In contrast, distributional properties of standard estimators are generally influenced by optional stopping. For instance, the sample average $(X_1+\cdots+X_N)/N$ is generally a biased estimator of $E_\theta X_1$. (See Problems 5.10 and 5.12 for examples.)

The exponential family (5.2) has an extra canonical parameter $-B(\theta)$, therefore sequential experiments usually lead to curved exponential families. The inverse binomial example is unusual in this regard, basically because the experiment is conducted so that $\sum_{i=1}^{N}X_i$ must be the fixed constant m.

5.3 Multinomial Distribution and Contingency Tables

The multinomial distribution is a generalization of the binomial distribution arising from n independent trials with outcomes in a finite set, $\{a_0,\dots,a_s\}$ say. Define vectors

$$e_0=\begin{pmatrix}1\\0\\0\\\vdots\\0\\0\end{pmatrix},\quad e_1=\begin{pmatrix}0\\1\\0\\\vdots\\0\\0\end{pmatrix},\ \dots,\quad e_s=\begin{pmatrix}0\\0\\0\\\vdots\\0\\1\end{pmatrix}$$

in $\mathbb{R}^{s+1}$, and take $Y_i=e_j$ if trial i has outcome a_j, $i=1,\dots,n$, $j=0,\dots,s$. Then $Y_1,\dots,Y_n$ are i.i.d. If p_j, $j=0,\dots,s$, is the chance of outcome a_j, then $P(Y_i=e_j)=p_j$. If we define

$$X=\begin{pmatrix}X_0\\X_1\\\vdots\\X_s\end{pmatrix}=\sum_{i=1}^{n}Y_i,$$

then X_j counts the number of trials with a_j as the outcome. By independence, the joint mass function of $Y_1,\dots,Y_n$ will be an n-fold product of success probabilities $p_0, \dots, p_s$. The number of times that p_j arises in this product will be the number of trials with outcome a_j, and so

$$P(Y_1 = y_1, \dots, Y_n = y_n) = \prod_{j=0}^{s} p_j^{x_j} = \exp\left[\sum_{j=0}^{s} x_j \log p_j\right],$$

where $x = x(y) = y_1 + \cdots + y_n$. Thus the joint distribution for $Y_1, \dots, Y_n$ form an $(s+1)$-parameter exponential family with canonical sufficient statistic X. But this family is not of full rank because $X_0 + \cdots + X_s = n$. Taking advantage of this constraint

$$P(Y_1 = y_1, \dots, Y_n = y_n) = \exp\left[\sum_{i=1}^{s} x_i \log(p_i/p_0) + n \log p_0\right],$$

which is a full rank s-parameter exponential family with complete sufficient statistic $(X_1, \dots, X_s)$. There is a one-to-one correspondence between this statistic and X, therefore X is also complete sufficient.

The distribution of X can be obtained from the distribution of Y as

$$P(X = x) = \sum_{(y_1,\dots,y_n):\sum_{i=1}^n y_i = x} P(Y = y). \tag{5.3}$$

The probabilities in this sum all equal $\prod_{j=1}^{s} p_j^{x_j}$, and so this common value must be multiplied by the number of ways the y_j can sum to x. This is equal to the number of ways of partitioning the set of trials $\{1, \dots, n\}$ into $s+1$ sets, the first with x_0 elements, the next with x_1 elements, and so on. This count is a *multinomial coefficient* given by

$$\binom{n}{x_0, \dots, x_s} = \frac{n!}{x_0! \times \cdots \times x_s!}.$$

This formula can be derived recursively. There are $\binom{n}{x_0}$ ways to choose the first set, then $\binom{n-x_0}{x_1}$ ways to choose the second set, and so on. The product of these binomial coefficients simplifies to the stated result. Using a multinomial coefficient to evaluate the sum in (5.3),

$$P(X_0 = x_0, \dots, X_s = x_s) = \binom{n}{x_0, \dots, x_s} p_0^{x_0} \times \cdots \times p_s^{x_s},$$

provided $x_0, \dots, x_s$ are nonnegative integers summing to n. This is the mass function for the *multinomial distribution*, and we write

$$X \sim \text{Multinomial}(p_0, \dots, p_s; n).$$

The marginal distribution of X_j, because X_j counts the number of trials with a_j as an outcome, is binomial with success probability p_j. Because X is complete sufficient, X_j/n is UMVU for p_j. Unbiased estimation of the product p_jp_k of two different success probabilities is more interesting as X_j and X_k are dependent. One unbiased estimator δ is the indicator that Y_1 is a_j and Y_2 is a_k. The chance that $X = x$ given $\delta = 1$ is a multinomial probability for $n-2$ trials with outcome a_j occurring $x_j - 1$ times and outcome a_k occurring $x_k - 1$ times. Therefore

$$\begin{aligned} P(\delta = 1|X = x) &= \frac{P(\delta = 1)P(X = x|\delta = 1)}{P(X = x)} \\ &= \frac{p_jp_k\binom{n-2}{x_0,\dots,x_j-1,\dots,x_k-1,\dots,x_s}p_0^{x_0}\times\cdots\times p_s^{x_s}/(p_jp_k)}{\binom{n}{x_0,\dots,x_s}p_0^{x_0}\times\cdots\times p_s^{x_s}} \\ &= \frac{x_jx_k}{n(n-1)}. \end{aligned}$$

Thus $X_jX_k/(n^2 - n)$ is UMVU for p_jp_k, $j \neq k$.

In applications, the success probabilities $p_0, \dots, p_s$ often satisfy additional constraints. In some cases this will lead to a full rank exponential family with fewer parameters, and in other cases it will lead to a curved exponential family. Here are two examples of the former possibility.

Example 5.5. Two-Way Contingency Tables. Consider a situation with n independent trials, but now for each trial two characteristics are observed: Characteristic A with possibilities $A_1, \dots, A_I$, and Characteristic B with possibilities $B_1, \dots, B_J$. Let N_{ij} denote the number of trials in which the combination A_iB_j is observed, and let p_{ij} denote the chance of A_iB_j on any given trial. Then

$$N = (N_{11}, N_{12}, \dots, N_{IJ}) \sim \text{Multinomial}(p_{11}, p_{12}, \dots, p_{IJ}; n).$$

These data and the sums

$$N_{i+} = \sum_{j=1}^{J} N_{ij}, \qquad i = 1, \dots, I,$$

and

$$N_{+j} = \sum_{i=1}^{I} N_{ij}, \qquad j = 1, \dots, J,$$

are often presented in a contingency table with the following form:

	B_1	$\cdots$	B_J	Total
A_1	N_{11}	$\cdots$	N_{1J}	N_{1+}
$\vdots$	$\vdots$		$\vdots$	$\vdots$
A_I	N_{I1}	$\cdots$	N_{IJ}	N_{I+}
Total	N_{+1}	$\cdots$	N_{+J}	n

If characteristics A and B are independent, then

$$p_{ij} = p_{i+}p_{+j}, \qquad i = 1, \dots, I, \quad j = 1, \dots, J,$$

where $p_{i+} = \sum_{j=1}^{J} p_{ij}$ is the chance of A_i, and $p_{+j} = \sum_{i=1}^{I} p_{ij}$ is the chance of B_j. With independence, the mass function of N can be written as

$$\binom{n}{n_{11}, \dots, n_{IJ}} \prod_{i=1}^{I} \prod_{j=1}^{J} (p_{i+}p_{+j})^{n_{ij}}.$$

Letting $n_{i+} = \sum_{j=1}^{J} n_{ij}$, $i = 1, \dots, I$, and $n_{+j} = \sum_{i=1}^{I} n_{ij}$, $j = 1, \dots, J$,

$$\prod_{i=1}^{I} \prod_{j=1}^{J} p_{i+}^{n_{ij}} = \prod_{i=1}^{I} p_{i+}^{n_{i+}} \text{ and } \prod_{i=1}^{I} \prod_{j=1}^{J} p_{+j}^{n_{ij}} = \prod_{j=1}^{J} p_{+j}^{n_{+j}}.$$

So the mass function of N can be written

$$\binom{n}{n_{11}, \dots, n_{IJ}} \prod_{i=1}^{I} p_{i+}^{n_{i+}} \prod_{j=1}^{J} p_{+j}^{n_{+j}}.$$

Using the constraints $\sum_{i=1}^{I} n_{i+} = \sum_{j=1}^{J} n_{+j} = n$ and $\sum_{i=1}^{I} p_{i+} = \sum_{j=1}^{J} p_{+j} = 1$, this mass function equals

$$\binom{n}{n_{11}, \dots, n_{IJ}} \exp\Biggl[\sum_{i=2}^{I} n_{i+} \log\left(\frac{p_{i+}}{p_{1+}}\right) + \sum_{j=2}^{J} n_{+j} \log\left(\frac{p_{+j}}{p_{+1}}\right) + n \log(p_{1+}p_{+1})\Biggr].$$

These mass functions form a full rank $(I+J-2)$-parameter exponential family with canonical sufficient statistic

$$(N_{2+}, \dots, N_{I+}, N_{+2}, \dots, N_{+J}).$$

The equivalent statistic

$$(N_{1+}, \dots, N_{I+}, N_{+1}, \dots, N_{+J})$$

is also complete sufficient. In this model, $N_{i+} \sim \text{Binomial}(n, p_{i+})$ and $N_{+j} \sim \text{Binomial}(n, p_{+j})$ are independent. So $\hat{p}_{i+} = N_{i+}/n$ and $\hat{p}_{+j} = N_{+j}/n$ are UMVU estimates of p_{i+} and p_{+j}, respectively, and $\hat{p}_{i+}\hat{p}_{+j}$ is the UMVU estimate of $p_{ij} = p_{i+}p_{+j}$.

Example 5.6. Tables with Conditional Independence. Suppose now that three characteristics, A, B, and C, are observed for each trial, with N_{ijk} the number of trials that result in combination $A_iB_jC_k$, and p_{ijk} the chance of this combination. Situations frequently arise in which it seems that characteristics A and B should be unrelated, but they are not independent because both are influenced by the third characteristic C. An appropriate model may be that A and B are conditionally independent given C. This leads naturally to the following constraints on the cell probabilities:

$$p_{ijk} = p_{++k}\frac{p_{i+k}}{p_{++k}}\frac{p_{+jk}}{p_{++k}}, \qquad i = 1,\dots,I,\ \ j = 1,\dots,J,\ \ k = 1,\dots,K,$$

where a "+" as a subscript indicates that the values for that subscript should be summed. Calculations similar to those for the previous example show that the mass functions with these constraints form a full rank $\bigl(K(I+J-1)-1\bigr)$-parameter exponential family with sufficient statistics N_{++k}, $k = 1,\dots,K$, N_{i+k}, $i = 1,\dots,I$, $k = 1,\dots,K$, and N_{+jk}, $j = 1,\dots,J$, $k = 1,\dots,K$.

5.4 Problems[2]

*1. Suppose X has a binomial distribution with m trials and success probability θ, Y has a binomial distribution with n trials and success probability θ^2, and X and Y are independent.
   a) Find a minimal sufficient statistic T.
   b) Show that T is not complete, providing a nontrivial function f with $E_\theta f(T) = 0$.

*2. Let X and Y be independent Bernoulli variables with $P(X = 1) = p$ and $P(Y = 1) = h(p)$ for some known function h.
   a) Show that the family of joint distributions is a curved exponential family unless
   $$h(p) = \frac{1}{1 + \exp\left\{a + b\log\frac{p}{1-p}\right\}}$$
   for some constants a and b.
   b) Give two functions h, one where (X, Y) is minimal but not complete, and one where (X, Y) is minimal and complete.

3. Let X and Y be independent Poisson variables.
   a) Suppose X has mean λ, and Y has mean λ^2. Do the joint mass functions form a curved two-parameter exponential family or a full rank one-parameter exponential family?
   b) Suppose instead X has mean λ, and Y has mean 2λ. Do the joint mass functions form a curved two-parameter exponential family, or a full rank one-parameter exponential family?

[2] Solutions to the starred problems are given at the back of the book.

4. Consider the two-sample problem with $X_1, \dots, X_m$ i.i.d. from $N(\mu_x, \sigma_x^2)$ and $Y_1, \dots, Y_n$ i.i.d. from $N(\mu_y, \sigma_y^2)$, and all $n + m$ variables mutually independent.
   a) Find the UMVU estimator for the ratio of variances, σ_x^2/σ_y^2.
   b) If the two variances are equal, $\sigma_x^2 = \sigma_y^2 = \sigma$, find the UMVU estimator of the normalized difference in means $(\mu_x - \mu_y)/\sigma$.
5. Consider the two-sample problem with $X_1, \dots, X_m$ i.i.d. from $N(\mu_x, \sigma^2)$ and $Y_1, \dots, Y_n$ i.i.d. from $N(\mu_y, \sigma^2)$, and all $n + m$ variables mutually independent. Fix $\alpha \in (0, 1)$ and define a parameter q so that $P(X_i > Y_i + q) = \alpha$. Find the UMVU of q.

*6. Two teams A and B play a series of games, stopping as soon as one of the teams has 4 wins. Assume that game outcomes are independent and that on any given game team A has a fixed chance θ of winning. Let X and Y denote the number of games won by the first and second team, respectively.
   a) Find the joint mass function for X and Y. Show that as θ varies these mass functions form a curved exponential family.
   b) Show that $T = (X, Y)$ is complete.
   c) Find a UMVU estimator of θ.

*7. Consider a sequential experiment in which observations are i.i.d. from a Poisson distribution with mean λ. If the first observation X is zero, the experiment stops, and if $X > 0$, a second observation Y is observed. Let $T = 0$ if $X = 0$, and let $T = 1 + X + Y$ if $X > 0$.
   a) Find the mass function for T.
   b) Show that T is minimal sufficient.
   c) Does this experiment give a curved two-parameter exponential family or full rank one-parameter exponential family?
   d) Is T a complete sufficient statistic? Hint: Write $e^\lambda E_\lambda g(T)$ as a power series in λ and derive equations for g setting coefficients for λ^x to zero.

8. Potential observations (X_1, Y_1), $(X_2, Y_2), \dots$ in a sequential experiment are i.i.d. The marginal distribution of X_i is Poisson with parameter λ, the marginal distribution of Y_i is Bernoulli with success probability $1/2$, and X_i and Y_i are independent. Suppose we continue observation, stopping the first time that $Y_i = 1$, so that the sample size is
$$N = \inf\{i : Y_i = 1\}.$$
   a) Show that the joint densities form an exponential family, and identify a minimal sufficient statistic. Is the family curved?
   b) Find two different unbiased estimators of λ, both functions of the minimal sufficient statistic. Is the minimal sufficient statistic complete?
9. Consider an experiment observing independent Bernoulli trials with unknown success probability $\theta \in (0, 1)$. Suppose we observe trial outcomes until there are two successes in a row.
   a) Find a minimal sufficient statistic.

b) Give a formula for the mass function of the minimal sufficient statistic.
c) Is the minimal sufficient statistic complete? If it is, explain why, and if it is not, find a nontrivial function with constant expectation.

10. Consider a sequential experiment in which X_1 and X_2 are independent exponential variables with failure rate λ. If $X_1 < 1$, sampling stops after the first observation; if not, the second variable X_2 is also sampled. So $N = 1$ if $X_1 < 1$ and $N = 2$ if $X_1 \geq 1$.
   a) Do the densities for this experiment form a curved two-parameter exponential family or a one-parameter exponential family?
   b) Find $E\overline{X}$, and compare this expectation with the mean $1/\lambda$ of the exponential distribution.
11. Suppose independent Bernoulli trials are performed until the number of successes and number of failures differ by 2. Let X denote the number of successes, Y the number of failures (so $|X - Y| = 2$), and θ the chance of success.
   a) Find the joint mass function for X and Y. Show that these mass functions form a curved exponential family with $T = (X, Y)$.
   b) Show that T is complete.
   c) Find the UMVU estimator for θ.
   d) Find $P(X > Y)$.
12. Consider a sequential experiment in which the potential observations $X_1, X_2, \ldots$ are i.i.d. from a geometric distribution with success probability $\theta \in (0, 1)$, so

$$P(X_i = x) = \theta(1 - \theta)^x, \qquad x = 0, 1, \ldots.$$

   The sampling rule calls for a single observation ($N = 1$) if $X_1 = 0$, and two observations ($N = 2$) if $X_1 \geq 1$. Define

$$T = \sum_{i=1}^{N} X_i.$$

   a) Do the densities for this experiment form a curved two-parameter exponential family or a one-parameter exponential family?
   b) Show that T is minimal sufficient.
   c) Find the mass function for T.
   d) Is T complete? Explain why or find a function g such that $g(T)$ has constant expectation.
   e) Find $E\overline{X}$. Is $\overline{X}$ an unbiased estimator of EX_1?
   f) Find the UMVU estimator of EX_1.

*13. Consider a single two-way contingency table and define $R = N_{11}+N_{12}$ (the first row sum), $C = N_{11} + N_{21}$ (the first column sum), and $D = N_{11} + N_{22}$ (the sum of the diagonal entries).
   a) Show that the joint mass function can be written as a full rank three-parameter exponential family with $T = (R, C, D)$ as the canonical sufficient statistic.

b) Relate the canonical parameter associated with D to the "cross-product ratio" α defined as $\alpha = p_{11}p_{22}/(p_{12}p_{21})$.
c) Suppose we observe m independent two-by-two contingency tables. Let n_i, $i = 1, \dots, m$, denote the trials for table i. Assume that cell probabilities for the tables may differ, but that the cross-product ratios for all m tables are all the same. Show that the joint mass functions form a full rank exponential family. Express the sufficient statistic as a function of the variables $R_1, \dots, R_m$, $C_1, \dots, C_m$, and $D_1, \dots, D_m$.

14. Consider a two-way contingency table with a multinomial distribution for the counts N_{ij} and with $I = J$. If the probabilities are symmetric, $p_{ij} = p_{ji}$, do the mass functions form a curved exponential family, or a full rank exponential family? With this constraint, identify a minimal sufficient statistic. Also, if possible, give UMVU estimators for the p_{ij}.
15. Let $(N_{11k}, N_{12k}, N_{21k}, N_{22k})$, $k = 1, \dots, n$, be independent two-by-two contingency tables. The kth table has a multinomial distribution with m trials and success probabilities

$$\left(\frac{1+\theta_k}{4}, \frac{1-\theta_k}{4}, \frac{1-\theta_k}{4}, \frac{1+\theta_k}{4}\right).$$

Note that θ_k can be viewed as a measure of dependence in table k. (If $\theta_k = 0$ there is independence in table k.) Consider a model in which

$$\log\left[\frac{1+\theta_k}{1-\theta_k}\right] = \alpha + \beta x_k, \qquad k = 1, \dots, m,$$

where α and β are unknown parameters, and $x_1, \dots, x_n$ are known constants. Show that the joint densities form an exponential family and identify a minimal sufficient statistic. Is this statistic complete?

*16. For an $I \times J$ contingency table with independence, the UMVU estimator of p_{ij} is $\hat{p}_{i+}\hat{p}_{+j} = N_{i+}N_{+j}/n^2$.
a) Determine the variance of this estimator, $\mathrm{Var}(\hat{p}_{i+}\hat{p}_{+j})$.
b) Find the UMVU estimator of the variance in (a).

17. In some applications the total count in a contingency table would most naturally be viewed as a random variable. In these cases, a Poisson model might be more natural than the multinomial model in the text.
a) Let $X_1, \dots, X_p$ be independent Poisson variables, and let λ_i denote the mean of X_i. Show that $T = X_1 + \cdots + X_p$ has a Poisson distribution, and find

$$P(X_1 = x_1, \dots, X_p = x_p | T = n),$$

the conditional mass function for X given $T = n$.
b) Consider a model for a two-by-two contingency table in which entries $N_{11}, \dots, N_{22}$ are independent Poisson variables, and let λ_{ij} denote the mean of N_{ij}. With the constraint $\lambda_{11}\lambda_{22} = \lambda_{12}\lambda_{21}$, do the joint mass functions for these counts form a curved four-parameter exponential family or a three-parameter exponential family?

18. Consider a two-way contingency table with a multinomial distribution for the counts N_{ij} with $I = J$. Assume that the cell probabilities p_{ij} are constrained to have the same marginal values,

$$p_{i+} = p_{+i}, \qquad i = 1, \dots, I.$$

a) If $I = 2$, find a minimal sufficient statistic T. Is T complete?
b) Find a minimal sufficient statistic T when $I = 3$. Is this statistic complete?
c) Suppose we add an additional constraint that the characteristics are independent, so

$$p_{ij} = p_{i+}p_{+j}, \qquad i = 1, \dots, I, \quad j = 1, \dots, I.$$

Give a minimal sufficient statistic when $I = 2$, and determine whether it is complete.